



2026-2027

Global Impact Academy

Networking Fundamentals

Global Impact Academy exists to cultivate a forward-thinking and community-driven (local, national, and global) learning environment where students develop a passion for STEM learning and its real-world applications. Global Impact Academy scholars will be productive, positive influencers of tomorrow, developing innovative solutions to the world's problems.

CONTACT INFORMATION

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ABOUT YOUR INSTRUCTOR

Hi, I'm Ms. T. Jones - an engineer turned educator, now in my 7th year of teaching. Before the classroom, I spent 12 years in the food industry in production and maintenance roles, and I bring that same real-world, problem-solving mindset to everything we do. My credentials include a B.S. in Biological and Agricultural Systems Engineering, a Master of Engineering in Systems Engineering, an Associate of Arts in Culinary Arts, a Post-Baccalaureate in Education, a Graduate Certificate in Cybersecurity, and an FCC Amateur Radio License (Technician Class) - and I am STEM and Gifted endorsed. I lead the Cybersecurity & Networking Division here at Global Impact Academy, where scholars join the GIA Technology Variance Agency as "agents" and take ownership of hands-on missions that build industry-ready skills. Outside of school, I love traveling, concerts, and cooking. A few of my favorite things - if you ever care to send a little encouragement - are Arden's Garden, Starbucks, and Chick-fil-A, plus everyday helps like gas and grocery gift cards. I look forward to partnering with you this year - please reach out anytime!

COURSE DESCRIPTION & OBJECTIVES

Networking Fundamentals (11.46100) is the Year-2 exploration course of the GIA Networking Pathway, taught through the GIA Technology Variance Agency (TVA) storyline: agents serve in Packet Patrol. Students master the physical layer, the OSI and TCP/IP models, protocols and services, IP addressing and subnetting, command-line diagnostics, WAN/routing, and network-security foundations. Concepts are made physical through micro:bit Networking activities, and a five-phase Data Center project runs all year, ending in a Community Planning Board presentation. The course is aligned to the Georgia CTAE Networking Fundamentals standards (IT-NF).

By the end of this course, students will be able to:

- Explain the OSI and TCP/IP models and the function of core protocols and services.
- Describe physical infrastructure and media, and build and verify cabling and connectivity.
- Apply IP addressing, subnetting, and IPv6 to design network architectures.
- Use command-line tools to configure and diagnose networks (Packet Tracer, Linux).
- Explain WAN technologies, routing, and network-security foundations.
- Design and document a data-center network through a multi-phase project.

STANDARDS OF LEARNING

- Georgia CTAE - Networking Fundamentals (IT-NF): IT-NF-1 & 2 (employability/portfolio), 3 (LAN fundamentals), 4 (OSI model), 5 (wired/wireless media), 6 (IP addressing/subnetting - power standard), 7 (command line), 8 (services), 9 (WAN/routing), 10 (infrastructure/security), 11 (CTSO).
- All-pathways foundations & employability standards (Unit 0); CTSO participation.
- Industry alignment: Cisco NetAcad and AWS Academy Cloud Foundations badges; Palo Alto PCCSA (stretch); Cisco Packet Tracer skills.
- No AP/IB/Dual Enrollment or EOPA - Year 2 Exploration course.

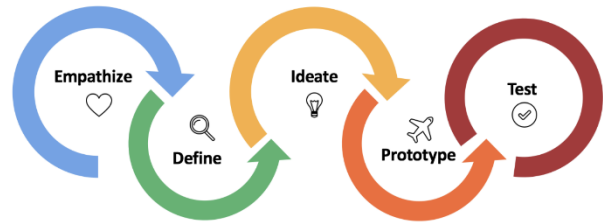
"Note to Parents and Guardians: Certain texts selected for this course may contain sensitive or controversial subject matter intended to support the development of specific academic skills and to promote thoughtful dialogue and deeper engagement with relevant topics, themes, or concepts. In recognition of your role in your student's education, you are entitled to review these materials and may request an alternative text for your student if you choose."

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COURSE PATHWAY CONNECTIONS

- ☐ Advanced Math
- ☐ Advanced Science
- ☐ Energy Systems
- ☐ Engineering
- ☐ Healthcare Sciences
 - ☐ Patient Care
 - ☐ Biotechnology Research and Development
 - ☐ Pharmacy
- ☐ Information Technology
 - ☐ Cybersecurity
 - ☐ Game Design
 - ☐ Computer Science

DESIGN THINKING PROCESS



TEXTBOOK & MATERIALS

ORGANIZATION REQUIREMENTS	SUPPLIES
Students will be required to keep their class files and assignments organized as follows: - o <u>OneDrive: class files organized by unit in Canvas/OneDrive.</u> - o <u>Notebook/Binder: a dedicated cumulative glossary notebook.</u> - o <u>Handouts:</u> Handouts will be kept in a pocket/prong folder to refer back to throughout the school year.	Glossary/composition notebook; headphones or earbuds; charged district device; pocket/prong folder for handouts; a USB flash drive. Business/business-casual attire is needed for presentations, FBLA events, and dress-up days.

ASSIGNMENTS & EVALUATION

Throughout this course, scholars will receive grades in the following categories and weights:

1st and 2nd Sem (Non-EOC) Major/Summative Assessments (55%), Minor Assessments (35%), and Practice Assignments (10%).

2nd Sem (EOC) Major/Summative Assessments (40%), Minor Assessments (30%), Practice Assignments (10%), and EOC (20%).

There is no End-of-Course Test (EOC) or EOPA for Networking Fundamentals; it is the Year-2 Exploration course. Grading follows the Non-EOC model (Major 55% · Minor 35% · Practice 10%) all year.

Department Grading Criteria: Types of Assignments Graded in Each Category	
Majors	Missions - Data Center PBL phase deliverables, Canvas Unit Tests, the Semester 1 Interim Review, and the Year 2 Capstone (Community Planning Board).
Minors	Minor Missions - project checkpoints (M1, M2) plus the cumulative Vocabulary Quiz (M3), which spirals at least five terms from prior units; from Unit 2, a weekly Port/Protocol Quiz.
Practices	Field Drills - Cisco Packet Tracer labs, micro:bit Networking builds, subnetting exercises, and the weekly Port/Protocol Quiz.

*These assignments are meant to 1) help the student gain mastery of Georgia K-12 Standards and 2) help the teacher understand where the scholar still needs support or enrichment.

Grading Practices

LATE/MISSING & MAKE-UP WORK POLICY [Please refer to Board Policy](#)

**Grading Absent Work:

- If a student misses an assignment due to being absent, they will be given the number of days equal to their absence to submit their work. If student turns in absent work by designated days equal to their absence, they are eligible to receive full credit.

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- If a student fails to submit work after days allotted for absence, the same point deduction policy will apply as the one below for “student present but fails to turn in work.”

****Student present but fails to turn in work:**

1. **Assignment Deadlines**
 - All **minor and practice assignments** are due **one week** after the date they are assigned.
 - **Major assignments** (e.g., assessments or projects) may have varying due dates, which will be clearly communicated at the time of assignment.
2. **Late Work Procedure**
 - Teacher may begin deducting points from late/missing assignments, assessments or tasks (maximum 25% deduction) to the overall final score. In other words, if the student submits an assignment, assessment, or task late and score 75%, the final score will reflect a 50%.
3. **Non-Submission**
 - If the assignment is **never submitted**, a **zero** will remain in the gradebook.
4. **Accommodations and Modifications**
 - Students with documented accommodations or modifications will have their needs met in accordance with their individualized learning plans.

****Major Assessment Recovery:**

- If a student scores below 75% on a major assessment, they will be given one retake opportunity to replace the grade. Before recovery is offered, the student must complete any missing work and/or attend tutoring services for reteaching and relearning. The highest grade that can be earned on the retake is 75%.

[Grading Policy Communication for Parents](#)

[Grading Policy Communication for Students](#)

EXTRA CREDIT POLICY

There is no formal chance for extra credit in this class. Scholars are encouraged to meet with their teacher to determine which assignment(s) they can revisit to regain credit.

ACADEMIC SUPPORTS

Here are some academic supports that may help your student be successful in this course:

Primary curriculum: Cyber.org Networking; micro:bit Networking (Nominet). **Simulation:** Cisco Packet Tracer; Intro to Linux; Raspberry Pi. **Data literacy:** Data Science for Everyone (DS4E) strands are threaded through the units - see the NF DS4E integration crosswalk. **Platforms:** Canvas (LMS), Spikeview. Fridays are reserved for enrichment & remediation.

Student Academic Habits Rubric [GIA Grading Rubric for Students Non-Academic Soft Skills.docx](#)

Non-academic skills will be reported in a separate section from academic performance. Feedback will be given to students and parents/guardians every 9 weeks on the progress report/report card.

CAREER TECHNICAL STUDENT ORGANIZATION (CTSO): FBLA

FBLA (Future Business Leaders of America) is the Career Technical Student Organization (CTSO) for the Digital Technology pathways at Global Impact Academy. All scholars are encouraged to join FBLA to attend conferences for professional development, network with peers and industry, and compete in technology and leadership events. Please note that FBLA membership requires annual dues, and there are additional costs for conferences, events, and field trips. Participation is optional but highly encouraged as a way to build professional skills and strengthen college and career readiness.

COURSE RESOURCES AND ABSENCES

Below you can review the unit themes/topics covered this year. The course comprises six instructional units (Units 1–6) plus a Unit 0 foundations week, aligned to the Georgia CTAE Networking Fundamentals standards. The five-phase Data Center project runs across the units, ending in the Year 2 Community Planning Board.

For **student absences**, students can find all class assignments and resources in CANVAS. Absent students making up any missing work will follow the **late work policy** outlined above. Please note that the sooner the student completes any missing work, the better their overall mastery and performance on class material and concepts.

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Units	Semester 1	Unit Dates (tentative)
1	Network Theory, Physical Infrastructure & Data Link	Aug 10 – Sep 4
2	OSI Model, Protocols & Network Addressing	Sep 8 – Oct 2
3	IP Addressing, Subnetting & IPv6	Oct 5 – Nov 6
	Semester 2	
4	Networking Services, Command Line & Diagnostics	Nov 9 – Dec 12
5	WAN Technologies, Routing & Security Foundations	Jan 5 – Feb 5
6	Network Design Capstone - Data Center PBL Final	Feb 8 – Apr 4

Grade Scale

Grade	Grade Points	100-Point Scale
A	4.0	90-100
B	3.0	80-89
C	2.0	70-79
F	0.0	<=69

Fulton County Grading Policies
 You may view all county policies on grading here:
[FCS Grading Policy](#)



Grading Expectations during 4.5-week, 9-week, 13.5-week and 18 week.

Minimum Grading Period-Weeks	Number of Majors	Number of Minors	Number of Practices
Up to 4 weeks	Actively Administering and Grading Assignments		
Grades are posted by the 4.5-week	1 major summative	1 minor formative	1 practice assignment
5-week to 8.5-week	Actively Administering and Grading Assignments		
Grades are posted by the 9-week	2 majors summative	3 minors formative	3 practices assignment
9.5-week to 13-week	Actively Administering and Grading Assignments		
Grades are posted by the 13.5-week	3 majors summative	4 minors formative	4 practices assignment
14-week to 17-week	Actively Administering and Grading Assignments		
Grades are posted by the 18-week	4 majors summative	6 minors formative	6 practices assignment
Maximum Grading Period-Weeks	6 - 7 majors *9 majors for ELA summative	12 – 13 minors formative	12 – 13 practices Assignments

Academic Probation [Academic Probation Information.pdf](#)

All students are on Academic monitoring during every 9-week grading period. If warranted, students will move to an Academic Warning Period for one 9-week grading period, Academic Probation Period for two 9-week grading periods, followed by an Eligibility Committee Status Review.

CELL PHONE POLICY –

When students enter the classroom, cell phones will be placed in a cell phone box provided by Global Impact Academy. The box will remain locked until the end of class, when students can retrieve their phones. Technology is a helpful tool, but it can also serve as a severe distraction for students.

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To help ensure that students receive the best possible education, GIA scholars are required to keep their phones in the cell phone lock box for the duration of class. The placement of the cell phone lock box and the dismissal procedures ensure that devices will remain safe and secure. Students may pick them up at the end of class.

In case of emergency, please call the front office of Global Impact Academy at **470-254-3905**, and students will be notified accordingly.

INSTRUCTOR COMMUNICATION & FEEDBACK

Communication email or phone calls will be responded to within **48** business hours (i.e., school days).

INFINITE CAMPUS

Scholars and their families can access grades and attendance records in real-time on Fulton County's Infinite Campus. The scholar's legal guardian must retrieve an access code from Student Services to log in. In-progress grades will appear in Infinite Campus so that students may see how they are performing. It is highly recommended that parents/guardians/students all keep a vigilant eye on Infinite Campus to stay on top of attendance, classwork, and class progress.

AI USE POLICY: [GIA Academic Dishonesty Procedure.pptx](#)

2026-27 School Year: Responsible AI Use Encouraged:

Students are encouraged to use AI tools responsibly in this course to assist with their learning and assignments. AI can serve as an aid to enhance understanding and interaction with course content, within the boundaries of academic integrity. All AI-generated content must be accurately cited, and students are responsible for the integrity and correctness of their submissions, including any content derived from AI tools. It is essential to critically evaluate AI outputs to ensure they meet academic standards. Unauthorized use of generative AI, without proper citation, is considered a violation of the Fulton County Schools Code of Conduct and may result in disciplinary action.

Assignment & Assessment AI Labels

Every assignment and assessment in this course is labeled with one of the following:

AI-Closed: AI assistants may not be used. The work reflects your own understanding. Used for in-class labs, skills checks, tests, and quizzes.

AI-Open: AI assistants are allowed as a learning tool. You disclose how you used AI and reflect on it. Used for homework, research, and learning new tools.

AI-Hybrid: The task has both. Each part states whether AI is allowed. Used for projects and capstones.

CYBER RANGE ACCESS & ACCEPTABLE USE POLICY

This course uses the CYBER.ORG Range, a professional cybersecurity lab environment. Before a student can receive a Range username and password, the student and their parent or guardian must review, sign, and return the Cyber.org Acceptable Use Policy (included in this course packet).

By signing, families agree that the student will use the Range only for authorized course activities, keep confidential any information learned during lab exercises, report any security vulnerabilities to the school without sharing them, and follow all district technology policies and applicable laws. Range accounts are monitored, and misuse may result in loss of access and disciplinary action.

ACADEMIC INTEGRITY [GIA Academic Dishonesty Procedure.pptx](#)

Global Impact Scholars are expected to hold themselves to the highest standard. As such, cheating will not be tolerated. Students who are caught cheating will receive a zero, marked as cheating in the gradebook, parent notification, and may also receive repercussions including loss of privileges such as extracurricular activities. If a student is caught cheating more than once, their counselor may be notified.

Examples of cheating include:

- Using AI to create something that was supposed to be done without any help or in an original manner.
- Copying or "borrowing" from another source and submitting it as one's own work.
- Seeking or accepting unauthorized assistance on tests, projects, or other assignments.
- Altering or forging grades, gradebooks, progress reports, report cards or other academic records
- Fabricating data, signatures, or resources.
- Providing or receiving test questions in advance without permission.
- Working collaboratively with other students when individual work is expected.

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Each student will read and sign off on agreement not to use AI or any resource or tool as an advantage.

[Updated Assessment Policies and Procedures Student Agreement.docx](#)

ACKNOWLEDGEMENT OF RECEIVING THIS SYLLABUS

After reviewing this syllabus, students will be asked to acknowledge receipt of the syllabus by submitting a digital signature on CANVAS.

Student Signature _____

Parent/Guardian Signature _____

Date _____

Career Opportunities

Networking Fundamentals (11.46100) · Networking Pathway

Global Impact Academy · Cybersecurity & Networking Division · GIA Technology Variance Agency

Georgia BCS Industry Certification — Standard 1, Item 1 (Career opportunities related to the course) · School Year 2026-2027 · Version 1.0

This addendum documents the career opportunities related to this course, for inclusion with the course syllabus. Networking Fundamentals develops the design, configuration, and support skills for computer-networking careers: the OSI and TCP/IP models, media and cabling, IP addressing and subnetting, command-line diagnostics, WAN and routing, and network-security foundations. It aligns to Cisco Networking Academy and CompTIA Network+.

Entry-Level Roles (with this course and its credential)

Role	What they do
Network Support Technician	Install, configure, and troubleshoot network connectivity.
Help Desk Technician (Network focus)	Resolve connectivity and access issues for users.
Structured Cabling Technician	Install and certify network cabling and hardware.

Advancement Roles (with continued education and experience)

Role	What they do
Network Administrator	Operate and maintain an organization's network.
Network Technician II	Configure switches, routers, and services.
Junior Network Engineer	Design and implement network solutions.

Credential Alignment

Cisco Networking Academy; AWS Academy Cloud Foundations; CompTIA Network+ N10-009 (stretch).

Place in the Pathway

Year 2 exploration course in the Networking pathway; continues to Networking Systems & Support (Year 3, launching SY2027-28), whose EOPA is ITS Networking (Certiport).

*Wage and outlook data for these occupations are available from the U.S. Bureau of Labor Statistics Occupational Outlook Handbook, O*NET OnLine, and the Georgia Department of Labor; teachers should cite current figures when presenting this addendum to students.*